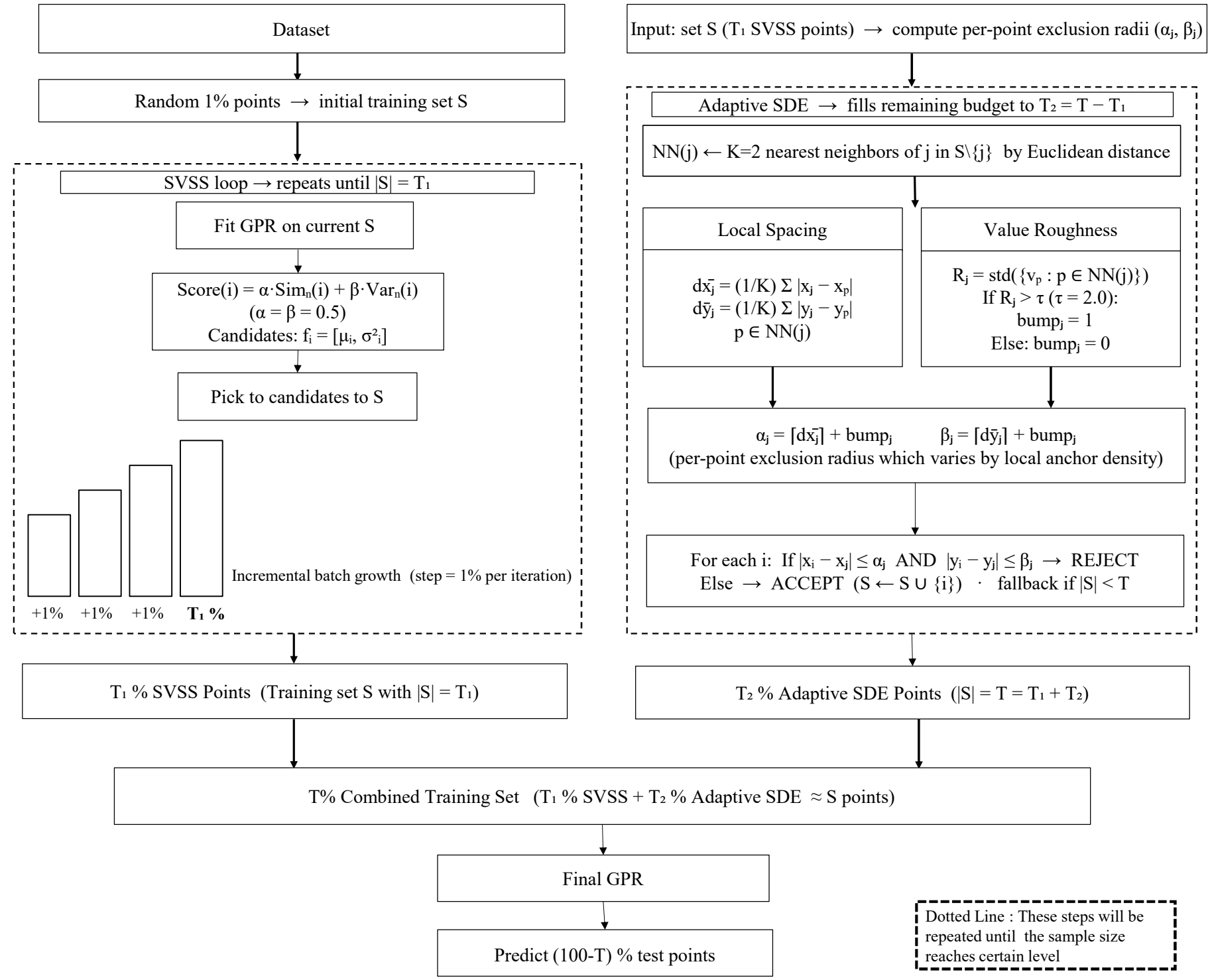

Phase 1 : Similarity-Variance based Sample Selection (SVSS)
Phase 2 : Adaptive Short Distance Elimination (ASDE)
Dataset
Random 1% points → initial training set S
SVSS loop → repeats until |S| = T₁
Fit GPR on current S
Score(i) = α·Sim_n(i) + β·Var_n(i)
(α = β = 0.5)
Candidates: f_i = [μ_i, σ²_i]
Pick to candidates to S
Incremental batch growth (step = 1% per iteration)
+1% +1% +1% T₁ %
T₁ % SVSS Points (Training set S with |S| = T₁)
Input: set S (T₁ SVSS points) → compute per-point exclusion radii (α_j, β_j)
Adaptive SDE → fills remaining budget to T₂ = T − T₁
NN(j) ← K=2 nearest neighbors of j in S\{j} by Euclidean distance
Local Spacing
dx̄_j = (1/K) Σ |x_j − x_p|
dȳ_j = (1/K) Σ |y_j − y_p|
p ∈ NN(j)
Value Roughness
R_j = std({v_p : p ∈ NN(j)})
If R_j > τ (τ = 2.0):
bump_j = 1
Else: bump_j = 0
α_j = ⌈dx̄_j⌉ + bump_j β_j = ⌈dȳ_j⌉ + bump_j
(per-point exclusion radius which varies by local anchor density)
For each i: If |x_i − x_j| ≤ α_j AND |y_i − y_j| ≤ β_j → REJECT
Else → ACCEPT (S ← S ∪ {i}) · fallback if |S| < T
T₂ % Adaptive SDE Points (|S| = T = T₁ + T₂)
T% Combined Training Set (T₁ % SVSS + T₂ % Adaptive SDE ≈ S points)
Final GPR
Predict (100-T) % test points
Dotted Line : These steps will be repeated until the sample size reaches certain level